Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Institute for Dynamic Systems and Control
IDSC
Institut für Dynamische Systeme und Regelungstechnik

Exam **August 5th, 2025**

Model Predictive Control (151-0660-00)

Prof. M. Zeilinger

Exam Supplementary Material

Please provide your student number and write your first and last name below.

Student number:

....

First and last name:

....

Exam Duration:	**120 Min**
Number of Questions:	**10**
Number of Points:	**100**
Permitted Aids:	**Two A4 sheets of paper (4 pages, handwritten or computer typed)**
Important:	No negative points will be given for incorrect answers.
	Answers should be given in the provided Answer booklet, or in the Question booklet, if required by the question.
	Use only the provided Answer booklet for your calculations; additional paper is available from the supervisors.
	For each multiple choice question (marked with **(MCQ)**) you are given four statements and you have to decide which of them are correct. For correct answers to all four statements the full number of points is allocated, for three correct answers half the number of points, and zero points otherwise.

Good luck!

S Supplementary Material

Definitions

Definitions used throughout the exam, which are for your reference and correspond to the definitions used in the lecture:

State sequence $X := \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_N \end{bmatrix}, \; x_i \in \mathbb{R}^n$

Input sequence $U := \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{bmatrix}, \; u_i \in \mathbb{R}^m$

Nominal state sequence $Z := \begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_N \end{bmatrix}, \; z_i \in \mathbb{R}^n$

Nominal input sequence $V := \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}, \; v_i \in \mathbb{R}^m$

One vector $\mathbb{1}_n := [1, 1, \ldots, 1]^T \in \mathbb{R}^n$

Identity matrix $\mathbb{I}_{n \times n} := \mathrm{diag}(\mathbb{1}_n) \in \mathbb{R}^{n \times n}$

Open interval $(-a, a) := \{x \in \mathbb{R} | -a < x < a\}$

Closed interval $[-a, a] := \{x \in \mathbb{R} | -a \leq x \leq a\}$

1-norm $\|x\|_1 = \|[x_1, \ldots, x_n]^T\|_1 := \sum_i |x_i|$

∞-norm $\|x\|_\infty = \|[x_1, \ldots, x_n]^T\|_\infty := \max_i(|x_i|)$

2-norm $\|x\|_2 = \|[x_1, \ldots, x_n]^T\|_2 := \sqrt{\sum_i x_i^2} = \sqrt{x^T x}$

Weighted 2-norm $\|x\|_Q = \|[x_1, \ldots, x_n]^T\|_Q := \sqrt{x^T Q x}$

Minkowski sum $\mathcal{A}_1 \oplus \mathcal{A}_2 = \{x_1 + x_2 | x_1 \in \mathcal{A}_1, \; x_2 \in \mathcal{A}_2\}$
(scalar case) $[a, b] \oplus [c, d] = [a + c, b + d]$

Pontryagin difference $\mathcal{A}_1 \ominus \mathcal{A}_2 = \{x_1 | x_1 + x_2 \in \mathcal{A}_1 \forall x_2 \in \mathcal{A}_2\}$
(scalar case) $[a, b] \ominus [c, d] = [a - c, b - d]$

Union $\mathcal{A}_1 \cup \mathcal{A}_2 = \{x | x \in \mathcal{A}_1 \text{ or } x \in \mathcal{A}_2\}$

Intersection $\mathcal{A}_1 \cap \mathcal{A}_2 = \{x | x \in \mathcal{A}_1, \, x \in \mathcal{A}_2\}$

Set difference $\mathcal{A}_1 \setminus \mathcal{A}_2 = \{x | x \in \mathcal{A}_1, \, x \notin \mathcal{A}_2\}$

- If not specified otherwise, an MPC control law refers to $\kappa(x(k)) = u_0^\star(x(k))$, i.e. the first element of the optimal input sequence computed by solving the associated optimization problem initialized at $x_0 = x(k)$.

S.1 Standard Nominal MPC Problem

Consider linear systems of the form

$$x(k+1) = Ax(k) + Bu(k). \tag{1}$$

The standard linear MPC control law $\kappa(x(k)) = u_0^\star(x(k))$ is defined by

$$
\begin{aligned}
P_{S.1}: \qquad J^\star(x(k)) = \min_{U} \quad & l_f(x_N) + \sum_{i=0}^{N-1} l(x_i, u_i) \\
\text{s.t.} \quad & x_{i+1} = Ax_i + Bu_i,\ i = 0, \ldots, N-1, && \text{(2a)} \\
& x_i \in \mathcal{X},\ i = 0, \ldots, N-1, && \text{(2b)} \\
& u_i \in \mathcal{U},\ i = 0, \ldots, N-1, && \text{(2c)} \\
& x_N \in \mathcal{X}_f, && \text{(2d)} \\
& x_0 = x(k). && \text{(2e)}
\end{aligned}
$$

Assumption S.1-1 The stage cost $l(x, u)$ is positive definite and only zero at the origin.

Assumption S.1-2

- The terminal set $\mathcal{X}_f$ is a positively invariant set for system (1) under the controller $\kappa_f(x)$:

$$Ax + B\kappa_f(x) \in \mathcal{X}_f, \quad \text{for all } x \in \mathcal{X}_f.$$

- All state and input constraints are satisfied in $\mathcal{X}_f$:

$$\mathcal{X}_f \subseteq \mathcal{X},\ \kappa_f(x) \in \mathcal{U}, \quad \text{for all } x \in \mathcal{X}_f.$$

Assumption S.1-3 The terminal cost is a continuous Lyapunov function in the terminal set $\mathcal{X}_f$ and satisfies:

$$l_f(Ax + B\kappa_f(x)) - l_f(x) \leq -l(x, \kappa_f(x)), \quad \text{for all } x \in \mathcal{X}_f.$$

S.2 Standard Tube-based MPC Problem

Consider uncertain linear systems of the form

$$x(k+1) = Ax(k) + Bu(k) + w(k) \tag{3}$$

with additive disturbances $w(k) \in \mathcal{W}$, where $\mathcal{W}$ is a compact set containing the origin.

The standard tube-based MPC control law $\kappa_{\text{tube}}(x(k)) = K_{\mathcal{E}}(x(k) - z_0^\star(x(k))) + v_0^\star(x(k))$ is defined by

$$P_{S.2}: \quad \tilde{J}^\star(x(k)) = \min_{V, z_0} \quad l_f(z_N) + \sum_{i=0}^{N-1} l(z_i, v_i)$$

$$\text{s.t.} \quad z_{i+1} = Az_i + Bv_i, \; i = 0, \dots, N-1, \tag{4a}$$

$$z_i \in \mathcal{X} \ominus \mathcal{E}, \; i = 0, \dots, N-1, \tag{4b}$$

$$v_i \in \mathcal{U} \ominus K_{\mathcal{E}}\mathcal{E}, \; i = 0, \dots, N-1, \tag{4c}$$

$$z_N \in \tilde{\mathcal{X}}_f, \tag{4d}$$

$$x(k) \in z_0 \oplus \mathcal{E}. \tag{4e}$$

Assumption S.2-1 The stage cost $l(z, v)$ is positive definite and only zero at the origin.

Assumption S.2-2

- The terminal set $\tilde{\mathcal{X}}_f$ is a positively invariant set for the nominal system $z(k+1) = Az(k) + Bv(k)$ under the controller $\tilde{\kappa}_f(z(k))$:

$$Az + B\tilde{\kappa}_f(z) \in \tilde{\mathcal{X}}_f, \quad \text{for all } z \in \tilde{\mathcal{X}}_f.$$

- All tightened state and input constraints are satisfied in $\tilde{\mathcal{X}}_f$:

$$\tilde{\mathcal{X}}_f \subseteq \mathcal{X} \ominus \mathcal{E}, \; \tilde{\kappa}_f(z) \in \mathcal{U} \ominus K_{\mathcal{E}}\mathcal{E}, \quad \text{for all } z \in \tilde{\mathcal{X}}_f.$$

Assumption S.2-3 The terminal cost is a continuous Lyapunov function in the terminal set $\tilde{\mathcal{X}}_f$ and satisfies:

$$l_f(Az + B\tilde{\kappa}_f(z)) - l_f(z) \leq -l(z, \tilde{\kappa}_f(z)), \quad \text{for all } z \in \tilde{\mathcal{X}}_f.$$

Assumption S.2-4 The set $\mathcal{E}$ is a robust invariant set for system (3) under $u(k) = K_{\mathcal{E}}x(k)$.

S.3 Standard Constraint-tightening MPC Problem

Consider uncertain linear systems of the form

$$x(k+1) = Ax(k) + Bu(k) + w(k) \tag{5}$$

with additive disturbances $w(k) \in \mathcal{W}$, where $\mathcal{W}$ is a compact set containing the origin.

The standard robust constraint-tightening MPC control law $\kappa_{\mathrm{ct}}(x(k)) = v_0^\star(x(k))$ is defined by

$$
\begin{aligned}
P_{S.3}: \qquad J_{\mathrm{ct}}^\star(x(k)) = \min_{V} \quad & l_f(z_N) + \sum_{i=0}^{N-1} l(z_i, v_i) \\
\text{s.t.} \quad & z_{i+1} = Az_i + Bv_i,\ i = 0, \ldots, N-1, && \text{(6a)}\\
& z_i \in \mathcal{X} \ominus \mathcal{F}_i,\ i = 0, \ldots, N-1, && \text{(6b)}\\
& v_i \in \mathcal{U} \ominus K_{\mathrm{ct}}\mathcal{F}_i,\ i = 0, \ldots, N-1, && \text{(6c)}\\
& z_N \in \mathcal{X}_f^{\mathrm{ct}} \ominus \mathcal{F}_N, && \text{(6d)}\\
& x(k) = z_0, && \text{(6e)}
\end{aligned}
$$

where $\mathcal{F}_0 = \{0\}$, $\mathcal{F}_i = \mathcal{W} \oplus (A + BK_{\mathrm{ct}})\mathcal{W} \oplus \ldots (A + BK_{\mathrm{ct}})^{i-1}\mathcal{W}$ for $i = 1 \ldots N$.

Assumption S.3-1 The stage cost $l(z, v)$ is positive definite and only zero at the origin.

Assumption S.3-2

- The terminal set $\mathcal{X}_f^{\mathrm{ct}}$ is a robust positively invariant set for system (5) under the terminal controller $\kappa_f^{\mathrm{ct}}(z(k)) = K_{\mathrm{ct}}z(k)$:

$$(A + BK_{\mathrm{ct}})z + w \in \mathcal{X}_f^{\mathrm{ct}}, \quad \text{for all } z \in \mathcal{X}_f^{\mathrm{ct}},\ \text{for all } w \in \mathcal{W}$$

- All state and input constraints are satisfied in $\mathcal{X}_f^{\mathrm{ct}}$:

$$\mathcal{X}_f^{\mathrm{ct}} \subseteq \mathcal{X}, \quad K_{\mathrm{ct}}\mathcal{X}_f^{\mathrm{ct}} \subseteq \mathcal{U}.$$